

CODE OF PRACTICE FOR THE ELECTRICITY (WIRING) REGULATIONS

Electrical and Mechanical Services Department
2025 Edition
(First Print)

Contents

	<i>Page</i>
Acknowledgements	11
Part I	
1. Introduction	12
2. Interpretation	13
3. Application	19
3A General Application of the CoP	20
3B Application to Category 2 Circuit	20
3C Exempted Fixed Electrical Installations	20
4. General Safety Requirements	21
4A General	22
4B Workmanship and Materials	23
4C Design, Construction, Installation and Protection	24
4D Identification, Maintenance, Inspection and Testing	25
4E Working Space	27
4F Switchroom/Substation	28
4G Safety Precautions for Work on Low Voltage Installation	29
4H Safety Precautions for Work on High Voltage Installation	32
4I General Safety Practices	39
5. Segregation of Circuit Categories	43
5A Circuit Category	44
5B Segregation of Category 1, 2 and 3 Circuits	45
5C Segregation of Category 4 Circuits and Circuits of Other Categories	46
5D Segregation of Circuits from Overhead Telecommunication Lines and Telephone Lines	47
6. Circuit Arrangement	49
6A Division of Installation into Circuits	50
6B Basic Requirements of Circuits	50
6C Ring Final Circuit Arrangement	53

	<i>Page</i>
6D Final Circuits Using 5A or 15A Socket Outlets to BS 546	53
6E Final Circuits Using 13A Socket Outlets to BS 1363	53
6F Final Circuits Using Universal Serial Bus (USB) Outlets to IEC 62368-1	54
6G Final Circuits Using 16A Industrial Socket Outlets to IEC 60309	55
6H Final Circuits Using 32A, 63A or 125A Industrial Socket Outlets to IEC 60309	55
6I Protection Requirement of High Voltage Circuit	56
7. Current Demand	62
7A Current Demand of a Circuit	63
7B Determination of Current Demand	63
8. Isolation and Switching	68
8A Provision of Isolation and Switching	69
8B Requirements of Isolation and Switching Devices	73
9. Overcurrent Protective Devices	77
9A General Requirements	78
9B Relation Between Circuit Conductors and Overcurrent Protective Devices	79
9C Breaking Capacity of Overcurrent Protective Devices	80
9D Position of Overcurrent Protective Devices	81
9E Other Requirements of Overcurrent Protective Devices	82
10. Neutral Conductor Protective Devices, Isolators and Switches	85
10A Linked Circuit Breakers, Linked Isolators and Linked Switches	86
10B Protection of Neutral Conductor	86
10C Single-pole Switch in Earthing Conductor	86
11. Earth Leakage and Earth Fault Currents	87
11A General	88
11B Basic Requirements	88
11C Protective Conductors	89
11D Earthing of Exposed Conductive Parts	91

	<i>Page</i>
11E Main Equipotential Bonding	92
11F Supplementary Bonding	94
11G Main Earthing Terminal	95
11H Earthing Conductors	96
11I Earth Fault Loop Impedance (Z_s)	97
11J Residual Current Devices	97
11K Earth Electrodes	99
12. Earthing Arrangement	109
12A General	110
12B Bonding Connections to the Point of Supply	110
12C Earth Electrode	111
13. Conductors, Joints and Connections	113
13A Selection and Sizing of Cable Conductors	114
13B Basic Protection	116
13C Joints and Connections	117
13D Other Requirements of Cables	118
14. Wiring Installation Enclosure	121
14A General Requirements	122
14B Steel Conduit System	123
14C Steel Trunking System	123
14D Plastic or PVC Conduit or Trunking	124
14E Cable Capacity of Enclosures	125
14F Enclosures of Busbar Trunking System and Powertrack System	126
14G Workmanship	126
15. Adverse Conditions Installation	130
15A Presence of Water (AD) or High Humidity (AB)	131
15B Ambient Temperature (AA)	132
15C Presence of Corrosive or Polluting Substance (AF)	134
15D Installation Exposed to Fire or Explosion	135
15E Impact (AG)	136
15F Vibration (AH)	137

	<i>Page</i>
15G Other Mechanical Stresses (AJ)	137
16. Overhead Line Installations	143
16A General	144
16B Installation of Overhead Lines	144
16C Joints of Overhead Lines	144
16D Service to Building	144
16E Conductor to Ground Clearance	144
16F Pole	145
16G Stay Wire	145
16H Carrier Wire	146
16I Earthing of Metallic Parts and Earth Leakage Protection	146
17. Display of Labels and Notices	147
17A Warning Notice for Substations and Switchrooms	148
17B Warning Notice for Connection of Earthing and Main Bonding Conductors	149
17C Warning Notice for Repair	149
17D Notice of Periodic Testing of Electrical Installations	149
17E Notice of Testing for RCCB / RCBO / AFDD	149
17F Warning Notice for Installation Having Both New and Old Cable Colours	150
17G Notice for Dual Power Supply	150
17H Warning Notice for Construction and Demolition Sites	150
18. Alterations and Additions	151
18A Requirements for Alterations or Additions to a Fixed Installation	152
18B Approval from the Electricity Supplier	153
19. First Inspection, Testing and Certification	154
19A Certification on Completion of an Installation	155
19B Work Completion Certificate	155
20. Periodic Inspection, Testing and Certification	157
20A Fixed Electrical Installations Specified in Regulation 20(1)	158

	<i>Page</i>
20B Fixed Electrical Installations Specified in Regulations 20(2), 20(3) and 20(4)	159
20C Periodic Test Certificate	159
21. Procedures for Inspection, Testing and Certification	161
21A Inspection of Low Voltage Installations	162
21B Testing of Low Voltage Installations	163
21C Inspection of High Voltage Installations	169
21D Testing of High Voltage Installations	169
21E Points to be Noted by Registered Electrical Workers	170
21F Power Suspension Arrangement for Periodic Inspection, Testing and Certification (PITC) Work on Main Switchboard connecting to Power Company's Transformer	171
22. Making and Keeping of Records	179
22A Keeping of Records by the Owner of an Electrical Installation that Requires Periodic Inspection, Testing and Certification	180
22B Making and Keeping of Records by a Registered Electrical Contractor	180
22C Types of Records	180
22D Checklists	181
23. (Reserved for Future Uses)	183
24. (Reserved for Future Uses)	183

	Part II	<i>Page</i>
25. General Workmanship		184
25A Wiring Installation Using Conduits		185
25B Wiring Installation Using Trunkings		187
25C Installation of Cables		188
25D Cable Joint and Cable Termination		191
25E Installation of Socket Outlets		195
26. Requirements for Specific Installations and Equipment		202
26A Domestic Installation and Appliance		205
26B Busbar Trunking Distribution System		211
26C Electric Motor		213
26D Supply Connection to Transformers		214
26E Supply Connection to Welding Sets		215
26F Installation of Fluorescent and Gaseous Discharge Lamps		215
26G Installation of Category 3 Circuits		215
26H High Voltage Discharge Lighting Installation (Neon Signs)		215
26I Lightning Protection Installation		219
26J Rising Mains Installation		219
26K Electrical Installation for Construction and Demolition Sites or Repair and Testing Purposes		220
26L Hot Air Saunas Installation		225
26M Swimming Pools and Fountains Installation		227
26N Installation in Restrictive Conductive Locations		234
26O Installation of Equipment Having High Earth Leakage Currents		235
26P Renewable Energy Power System		238
26Q Temporary Electrical Installation for Exhibitions, Shows, Stands and Festive Lighting		242
26R Electric Heating Systems installed in Building Structures		244
26S Charging Facilities for Electric Vehicles		245
26T Installation for Modular Integrated Construction / Multi-trade Integrated Mechanical, Electrical and Plumbing		249

	<i>Page</i>
26U Innovation & Technology Initiatives for Switchboard and Switchroom Applications	252
26V Battery Energy Storage System	254
Appendices	272
1. (Reserved for Future Uses)	273
2. (Reserved for Future Uses)	273
3. (Reserved for Future Uses)	273
4. (Reserved for Future Uses)	273
5. Rating Factors for Sizing of Cable Conductors	273
6. Current Carrying Capacities and Voltage Drop Tables for PVC Insulated and XLPE Insulated Cables	279
7. Typical Methods of Installation of Cables	297
8. Graphical Symbols for Electrical Diagrams	301
9. (Reserved for Future Uses)	309
10. Degree of Protection Provided by Enclosures (IP Code)	309
11. Forms of Internal Separations for Switchgear Assemblies	311
12. (Reserved for Future Uses)	313
13. Sample of Schedule of Test Results for Electrical Wiring / Schedule of Residual Currents Measurement for Residential Flat and Checklists	314
14. References	361
15. Safety for Live Work	369
16. Sample of Permit-To-Work (LV/HV) / Sanction-For-Test (HV)	372
17. Sample of HV Enclosure Log Book / HV Padlock Movement Log Book	378
18. New Cable Colour Code for Fixed Electrical Installations- Installation Guidelines	380
19. Sample of Task-specific Risk Assessment Form for Work inside False Ceiling	388
Index	389

- (c) Flexible sheet heating elements should comply with the requirements of IEC 60335-2-96, and heating cables should comply with the requirements of IEC 60800.
- (d) The manufacturer's installation instructions should be followed.
- (e) A heating cable intended for laying directly in soil, concrete, cement screed or other materials used for road and building construction should be:
 - (i) capable of withstanding mechanical damage under the conditions that can reasonably be expected to prevail during its installation; and
 - (ii) constructed of material that will be resistant to damage from dampness and/or corrosion under normal conditions of service.
- (f) A heating cable laid directly in soil, a road or the structure of a building should be installed so that it:
 - (i) is completely embedded in the substance it is intended to heat;
 - (ii) does not suffer damage in the event of movement normally to be expected in it or in the substance in which it is embedded; and
 - (iii) complies in all respects with the manufacturer's instructions and recommendations.

26S Charging Facilities for Electric Vehicles

- (1) General
 - (a) The particular requirements of this Code should apply to the installation of charging facilities for electric vehicles (EV).
 - (b) Each final circuit of EV charging facility shall be installed as a separate radial circuit of the fixed electrical installation, protected by high breaking capacity (HBC) fuse or miniature circuit breaker (MCB) of suitable rating.
 - (c) Diversity may be allowed for a dedicated distribution circuit supplying multiple electric vehicle charging points if load control is available.
- (2) Classification of charging mode
 - (a) **Mode 1** – In Mode 1 charging, connection of the EV to the AC supply network utilizes a standardized BS 1363 socket outlet and a charging cable without communication function to the on-board charger of the EV.

- (b) **Mode 2** – In Mode 2 charging, an in-cable control box is incorporated into the charging cable assembly. The provision of fixed electrical installation for charging facility is similar to that for Mode 1 except that the final circuit, protective device and socket outlet shall be of a suitable rating to cater for the higher level of charging current not exceeding 32A.
- (c) **Mode 3** – In Mode 3 charging, connection of the EV to the AC supply network utilizes a dedicated EV supply equipment (EVSE) where the control pilot function extends to control equipment in the EVSE, permanently connected to the AC supply network. Subject to the power rating of the on-board charger of an EV, Mode 3 charging can deliver a higher charging current (e.g. 220V/32A, 380V/32A, 380V/63A).
- (d) **Mode 4** – In Mode 4 charging, connection of the EV to the AC supply network utilizes an off-board charger where the control pilot function extends to equipment permanently connected to the AC supply. In this charging mode, either single-phase or three-phase AC is converted to DC within the EV charging equipment. The resulting DC is supplied to the EV via a charging cable that is tethered to the EV charging equipment.

(3) Selection and erection of installation

- (a) The EV charging installation should be selected and erected to ensure safe operation and ease of maintenance at all times. The EV charging installation should be designed and installed in accordance with IEC 61851 or equivalent. In addition, if the EV charging installation is designed for outdoor use, the equipment shall be selected with a degree of protection of at least IP44 in accordance with IEC 60529 to protect against water splashes (AD4) and the ingress of very small objects (AE3) respectively.
- (b) Socket Outlets and Connectors
 - (i) the socket outlets or connectors shall comply with the following standards or equivalent;
 - in Mode 1 charging – BS 1363;
 - in Mode 2 charging – IEC 60309 which is interlocked and classified to prevent the socket contacts being live when accessible ;
 - in Mode 3 charging – IEC 62196;

- (ii) In Mode 3 and 4 charging, an electrical or mechanical system shall be provided to prevent the plugging/unplugging of the plug unless the socket-outlet or the vehicle connector has been switch off from the supply.
 - (iii) one socket outlet or connector shall supply only one EV.
 - (iv) The socket outlet shall, in general, be installed at a height of about 1.2m above finished floor level for easy access and be positioned close to the parking space.
- (4) Protection for safety
 - (a) Protection against electric shock
 - (i) The protective measures of obstacles and placing out of reach are not permitted.
 - (ii) The protective measures of non-conducting locations and earth-free local equipotential bonding are not permitted.
 - (b) Protection against danger
 - (i) The EV charging facility shall be installed in a position to minimize the likelihood of vehicle impact damage. EV charging facility installed in public area and car park shall be protected against a minimum impact severity of AG3, or a minimum degree of protection against external mechanical impact of IK08 in accordance with the requirements of BS EN 62262.
 - (c) Devices for isolation and switching
 - (i) A means of isolating the EV charging equipment circuit shall be provided. This isolating device should be fixed to a maximum height of 2 m above finished floor level and accessible without using tools for maintenance purposes and shall be suitably identified by marking and/or labelling.
 - (ii) A current breaking device (an ON/OFF switch or others) (see Code 8B) shall be provided at the upstream of the socket outlet at the charging facility for switching on after plugging and switching off before unplugging the charging cable assembly.
 - (d) Fault protection
 - (i) Except for circuits using the protective measure of electrical separation, each charging point shall be protected by its own RCD of at least Type A, having the characteristic specified in Code 11J.

- (ii) Each charging point incorporating a socket outlet or connector complying with the IEC 62196 series, protective measures against DC fault current shall be taken, except where provided by the EV charging equipment. The appropriate measures, for each connection point, shall be as follows:
 - RCD Type B; or
 - RCD Type A in conjunction with a residual direct current detecting device (RDC-DD) complying with BS IEC 62955 as appropriate to the nature of the residual and superimposed currents and the recommendations of the manufacturer of the charging equipment.
- (iii) An RCD Type A in conjunction with an RDC-DD can be arranged with the RDC-DD inside the EV charging equipment and the Type A RCD upstream in either the charging equipment or the installation.
- (iv) DC EV charging installation for Mode 4 charging should comply with the requirements specific to DC EV charging facilities given under Part 23 of the IEC 61851 (i.e. IEC 61851-23).
- (v) For isolated DC EV charging equipment complying with IEC 61851-23, the DC circuit on output side of the EV charging equipment is electrically separated from AC circuit on power system side, which is acceptable as a means of protective measure by electrical separation. Protection by an RCD is optional for such charging equipment.
- (vi) For any non-isolated DC EV charging equipment where the DC circuit on output side of the EV charging equipment is not electrically separated from AC circuit on power system side, an RCD of at least Type A having the characteristic specified in Code 11J of the CoP shall be provided.
- (vii) DC EV charging equipment in compliance with IEC 61851-23 is required to be compatible with an RCD Type A in the installation as specified in Clause 7.6 of the standard (i.e. an RCD Type A, when installed on the AC supply circuit upstream of the charging equipment, shall be able to use together with the charging equipment and function safely as intended). Therefore, an RCD Type B is not required for DC EV charging equipment in compliance with IEC 61851-23.

- (viii) The classification of DC EV charging equipment and its conformity for protection against electric shock in compliance with IEC 61851-23 should be checked against the type test certificate and relevant test report of the equipment. Relevant clauses of IEC 61851-23 may include but not limited to the following:
- Clause 6.101 on classification of DC EV charging equipment & systems;
 - Clause 7 on protection against electric shock; and
 - Annexes AA.3.1, BB.2 or CC.4.1 on requirements for the isolated DC EV charging equipment for protection against electric shock.
- (ix) AFDDs shall be provided for final circuits of EV charging facilities as specified in Code 6B.

26T Installation for Modular Integrated Construction / Multi-trade Integrated Mechanical, Electrical and Plumbing

(1) Scope

The particular requirements of this Code should apply to fixed electrical installations for modular integrated construction (MiC) and Multi-trade Integrated Mechanical, Electrical and Plumbing (MiMEP). MiC refers to a construction whereby free-standing integrated modules (completed with finishes, fixtures and fittings) are manufactured in a prefabrication factory and then transported to site for installation in a building. MiMEP refers to the integration of multi-trade building services components into a single volumetric assembly of prefabricated modules, manufactured off-site in a workshop, then transported to the site for connection of modules to complete various trades of building services installations.

(2) Certification of electrical work

- (a) When a registered electrical contractor is employed to carry out the design of fixed electrical installation including the wiring installation with MiC/ MiMEP method, a registered electrical worker employed by this registered electrical contractor shall certify the design of fixed electrical installation and this registered electrical contractor shall endorse the certificate (i.e. Part 1 of Work Completion Certificate) to confirm the fixed electrical installation has been designed in accordance with the Electricity Ordinance.